```
%pip install matplotlib pandas numpy seaborn
```

```
Collecting matplotlib
  Downloading matplotlib-3.11.2-cp314-cp314-win_amd64.whl.metadata (80
kB)
Collecting pandas
  Downloading pandas-3.1.0-cp314-cp314-win_amd64.whl.metadata (18 kB)
Collecting numpy
  Downloading numpy-2.5.4-cp314-cp314-win_amd64.whl.metadata (6.6 kB)
Collecting seaborn
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.4.0-cp314-cp314-win_amd64.whl.metadata (3.8
kB)
Collecting cycler>=0.10 (from matplotlib)
  Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Downloading fonttools-4.68.0-cp314-cp314-win_amd64.whl.metadata (142
kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.5.1-cp314-cp314-win_amd64.whl.metadata (5.2
kB)
Requirement already satisfied: packaging>=20.0 in .\AppData\Local\
Programs\Python\Python314\Lib\site-packages (from matplotlib) (26.3)
Collecting pillow>=9 (from matplotlib)
  Downloading pillow-13.0.0-cp314-cp314-win_amd64.whl.metadata (9.3
kB)
Collecting pyparsing>=3 (from matplotlib)
  Using cached pyparsing-3.3.3-py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: python-dateutil>=2.7 in .\AppData\
Local\Programs\Python\Python314\Lib\site-packages (from matplotlib)
(4.0.0)
Requirement already satisfied: tzdata in .\AppData\Local\Programs\
Python\Python314\Lib\site-packages (from pandas) (2026.6)
Downloading matplotlib-3.11.2-cp314-cp314-win_amd64.whl (9.5 MB)
   ---------------------------------------- 0.0/9.5 MB ? eta -:--:--
   - -------------------------------------- 0.3/9.5 MB ? eta -:--:--
   --- ------------------------------------ 0.8/9.5 MB 3.0 MB/s eta
0:00:03
   ----- ---------------------------------- 1.3/9.5 MB 2.5 MB/s eta
0:00:04
   -------- ------------------------------- 2.1/9.5 MB 2.6 MB/s eta
0:00:03
   ----------- ---------------------------- 2.6/9.5 MB 2.8 MB/s eta
0:00:03
   --------------- ------------------------ 3.7/9.5 MB 3.1 MB/s eta
0:00:02
   ------------------ --------------------- 4.5/9.5 MB 3.3 MB/s eta
0:00:02
   ----------------------- ---------------- 5.5/9.5 MB 3.5 MB/s eta
```

```
0:00:02
   --------------------------- ------------ 6.6/9.5 MB 3.6 MB/s eta
0:00:01
   ------------------------------- -------- 7.6/9.5 MB 3.7 MB/s eta
0:00:01
   ------------------------------------ --- 8.7/9.5 MB 3.9 MB/s eta
0:00:01
   ---------------------------------------  9.4/9.5 MB 4.0 MB/s eta
0:00:01
   ---------------------------------------- 9.5/9.5 MB 3.9 MB/s
0:00:02
Downloading pandas-3.1.0-cp314-cp314-win_amd64.whl (10.9 MB)
   ---------------------------------------- 0.0/10.9 MB ? eta -:--:--
   --- ------------------------------------ 1.0/10.9 MB 6.2 MB/s eta
0:00:02
   ------- -------------------------------- 2.1/10.9 MB 5.1 MB/s eta
0:00:02
   ---------- ----------------------------- 2.9/10.9 MB 4.7 MB/s eta
0:00:02
   -------------- ------------------------- 3.9/10.9 MB 4.8 MB/s eta
0:00:02
   ------------------ --------------------- 5.0/10.9 MB 4.7 MB/s eta
0:00:02
   --------------------- ------------------ 5.8/10.9 MB 4.8 MB/s eta
0:00:02
   ------------------------- -------------- 6.8/10.9 MB 4.9 MB/s eta
0:00:01
   ----------------------------- ---------- 8.1/10.9 MB 4.9 MB/s eta
0:00:01
   -------------------------------- ------- 8.9/10.9 MB 4.9 MB/s eta
0:00:01
   ------------------------------------ --- 10.0/10.9 MB 4.8 MB/s eta
0:00:01
   ---------------------------------------  10.7/10.9 MB 4.9 MB/s eta
0:00:01
   ---------------------------------------- 10.9/10.9 MB 4.5 MB/s
0:00:02
Downloading numpy-2.5.4-cp314-cp314-win_amd64.whl (12.7 MB)
   ---------------------------------------- 0.0/12.7 MB ? eta -:--:--
   ---------------------------------------- 0.0/12.7 MB ? eta -:--:--
   -- ------------------------------------- 0.8/12.7 MB 5.3 MB/s eta
0:00:03
   ------- -------------------------------- 2.4/12.7 MB 5.6 MB/s eta
0:00:02
   --------- ------------------------------ 3.1/12.7 MB 5.2 MB/s eta
0:00:02
   -------------- ------------------------- 4.5/12.7 MB 5.3 MB/s eta
0:00:02
   ------------------ --------------------- 5.8/12.7 MB 5.5 MB/s eta
```

```
0:00:02
   ----------------------- ---------------- 7.3/12.7 MB 5.7 MB/s eta
0:00:01
   --------------------------- ------------ 8.7/12.7 MB 5.8 MB/s eta
0:00:01
   ---------------------------- ----------- 9.2/12.7 MB 5.9 MB/s eta
0:00:01
   --------------------------------- ------ 10.5/12.7 MB 5.5 MB/s eta
0:00:01
   ------------------------------------- -- 11.8/12.7 MB 5.6 MB/s eta
0:00:01
   ---------------------------------------  12.6/12.7 MB 5.5 MB/s eta
0:00:01
   ---------------------------------------- 12.7/12.7 MB 5.3 MB/s
0:00:02
Downloading seaborn-0.13.2-py3-none-any.whl (294 kB)
Downloading contourpy-1.4.0-cp314-cp314-win_amd64.whl (240 kB)
Using cached cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.68.0-cp314-cp314-win_amd64.whl (2.5 MB)
   ---------------------------------------- 0.0/2.5 MB ? eta -:--:--
   ------------ --------------------------- 0.8/2.5 MB 5.8 MB/s eta
0:00:01
   ---------------------------- ----------- 1.8/2.5 MB 4.8 MB/s eta
0:00:01
   ---------------------------------------- 2.5/2.5 MB 4.4 MB/s
0:00:00
Downloading kiwisolver-1.5.1-cp314-cp314-win_amd64.whl (72 kB)
Downloading pillow-13.0.0-cp314-cp314-win_amd64.whl (7.3 MB)
   ---------------------------------------- 0.0/7.3 MB ? eta -:--:--
   ----- ---------------------------------- 1.0/7.3 MB 6.1 MB/s eta
0:00:02
   ---------- ----------------------------- 1.8/7.3 MB 6.0 MB/s eta
0:00:01
   ------------ --------------------------- 2.4/7.3 MB 3.8 MB/s eta
0:00:02
   ------------------ --------------------- 3.4/7.3 MB 4.3 MB/s eta
0:00:01
   ------------------------ --------------- 4.5/7.3 MB 4.2 MB/s eta
0:00:01
   ------------------------- -------------- 4.7/7.3 MB 4.0 MB/s eta
0:00:01
   --------------------------- ------------ 5.0/7.3 MB 3.4 MB/s eta
0:00:01
   ------------------------------- -------- 5.8/7.3 MB 3.6 MB/s eta
0:00:01
   ---------------------------------- ----- 6.3/7.3 MB 3.4 MB/s eta
0:00:01
   -------------------------------------- - 7.1/7.3 MB 3.4 MB/s eta
0:00:01
```

```
   ---------------------------------------- 7.3/7.3 MB 3.2 MB/s
0:00:02
Using cached pyparsing-3.3.3-py3-none-any.whl (126 kB)
Installing collected packages: pyparsing, pillow, numpy, kiwisolver,
fonttools, cycler, pandas, contourpy, matplotlib, seaborn

   ----------------------------------------  0/10 [pyparsing]
   ----------------------------------------  0/10 [pyparsing]
   ---- -----------------------------------  1/10 [pillow]
   ---- -----------------------------------  1/10 [pillow]
   ---- -----------------------------------  1/10 [pillow]
   ---- -----------------------------------  1/10 [pillow]
   ---- -----------------------------------  1/10 [pillow]
   ---- -----------------------------------  1/10 [pillow]
   ---- -----------------------------------  1/10 [pillow]
   ---- -----------------------------------  1/10 [pillow]
   ---- -----------------------------------  1/10 [pillow]
   ---- -----------------------------------  1/10 [pillow]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- -------------------------------  2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   -------- ------------------------------- 2/10 [numpy]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- ----------------------- 4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   ---------------- -----------------------  4/10 [fonttools]
   -------------------- -------------------  5/10 [cycler]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
```

```
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
    ------------------------ ---------------  6/10 [pandas]
```

```
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ------------------------ ---------------  6/10 [pandas]
   ---------------------------- -----------  7/10 [contourpy]
   ---------------------------- -----------  7/10 [contourpy]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   -------------------------------- -------  8/10 [matplotlib]
   ------------------------------------ ---  9/10 [seaborn]
   ------------------------------------ ---  9/10 [seaborn]
   ------------------------------------ ---  9/10 [seaborn]
   ------------------------------------ ---  9/10 [seaborn]
   ------------------------------------ ---  9/10 [seaborn]
   ------------------------------------ ---  9/10 [seaborn]
   ---------------------------------------- 10/10 [seaborn]

Successfully installed contourpy-1.4.0 cycler-0.12.1 fonttools-4.68.0
kiwisolver-1.5.1 matplotlib-3.11.2 numpy-2.5.4 pandas-3.1.0 pillow-
13.0.0 pyparsing-3.3.3 seaborn-0.13.2
Note: you may need to restart the kernel to use updated packages.


[notice] A new release of pip is available: 26.2.2 -> 26.3
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```python
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import seaborn as sns
%matplotlib inline
```

```python
student=pd.read_csv("C:/Users/VAISHNAVI/Downloads/archive.zip")
student
```

```
      StudentID  Age  Gender  Ethnicity  ParentalEducation
StudyTimeWeekly  \
0          1001   17       1          0                  2
19.833723
1          1002   18       0          0                  1
15.408756
2          1003   15       0          2                  3
4.210570
3          1004   17       1          0                  3
```

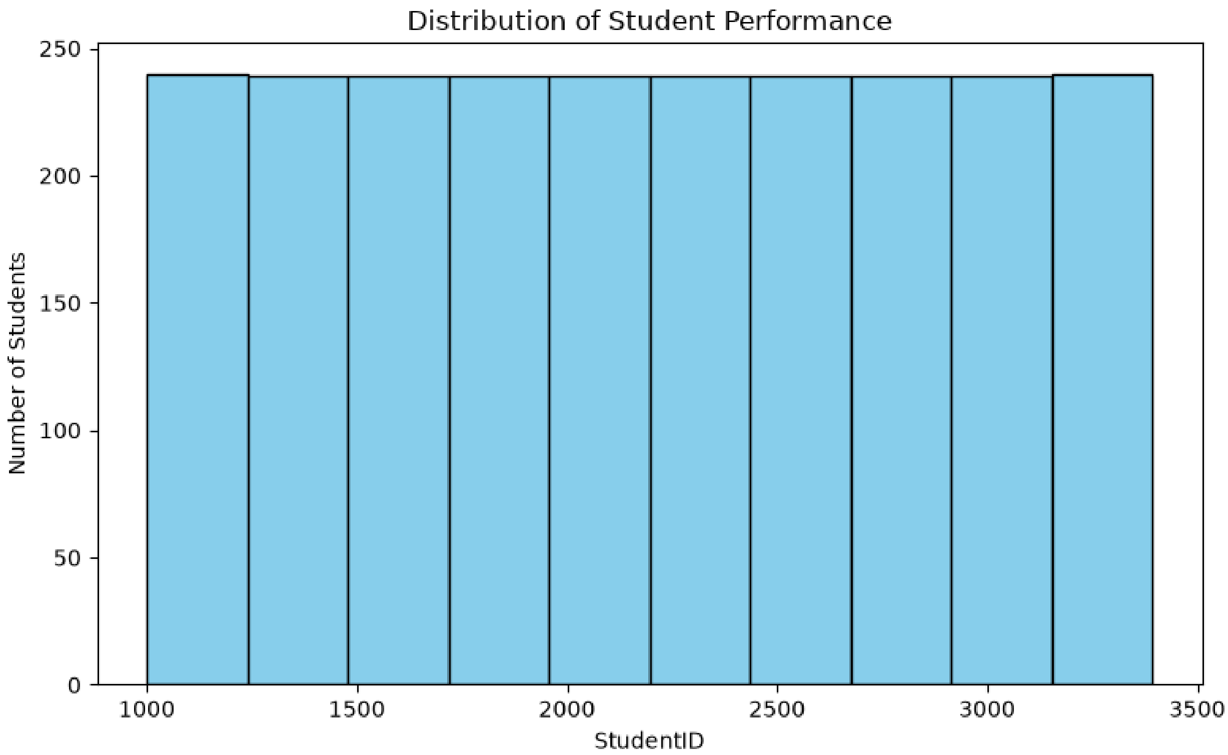
```
10.028829
4          1005   17       1          0                  2
4.672495
...         ...  ...     ...        ...                ...
...
2387       3388   18       1          0                  3
10.680555
2388       3389   17       0          0                  1
7.583217
2389       3390   16       1          0                  2
6.805500
2390       3391   16       1          1                  0
12.416653
2391       3392   16       1          0                  2
17.819907

      Absences  Tutoring  ParentalSupport  Extracurricular  Sports
Music  \
0            7         1                2                0       0
1
1            0         0                1                0       0
0
2           26         0                2                0       0
0
3           14         0                3                1       0
0
4           17         1                3                0       0
0
...        ...       ...              ...              ...     ...
...
2387         2         0                4                1       0
0
2388         4         1                4                0       1
0
2389        20         0                2                0       0
0
2390        17         0                2                0       1
1
2391        13         0                2                0       0
0

      Volunteering       GPA  GradeClass
0                0  2.929196         2.0
1                0  3.042915         1.0
2                0  0.112602         4.0
3                0  2.054218         3.0
4                0  1.288061         4.0
...            ...       ...         ...
2387             0  3.455509         0.0
```

```
2388              0  3.279150         4.0
2389              1  1.142333         2.0
2390              0  1.803297         1.0
2391              1  2.140014         1.0

[2392 rows x 15 columns]

student.info()

<class 'pandas.DataFrame'>
RangeIndex: 2392 entries, 0 to 2391
Data columns (total 15 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   StudentID          2392 non-null   int64  
 1   Age                2392 non-null   int64  
 2   Gender             2392 non-null   int64  
 3   Ethnicity          2392 non-null   int64  
 4   ParentalEducation  2392 non-null   int64  
 5   StudyTimeWeekly    2392 non-null   float64
 6   Absences           2392 non-null   int64  
 7   Tutoring           2392 non-null   int64  
 8   ParentalSupport    2392 non-null   int64  
 9   Extracurricular    2392 non-null   int64  
 10  Sports             2392 non-null   int64  
 11  Music              2392 non-null   int64  
 12  Volunteering       2392 non-null   int64  
 13  GPA                2392 non-null   float64
 14  GradeClass         2392 non-null   float64
dtypes: float64(3), int64(12)
memory usage: 280.4 KB

student.head()

   StudentID  Age  Gender  Ethnicity  ParentalEducation
StudyTimeWeekly  \
0       1001   17       1          0                  2
19.833723
1       1002   18       0          0                  1
15.408756
2       1003   15       0          2                  3
4.210570
3       1004   17       1          0                  3
10.028829
4       1005   17       1          0                  2
4.672495

   Absences  Tutoring  ParentalSupport  Extracurricular  Sports  Music
\
0         7         1                2                0       0      1
```

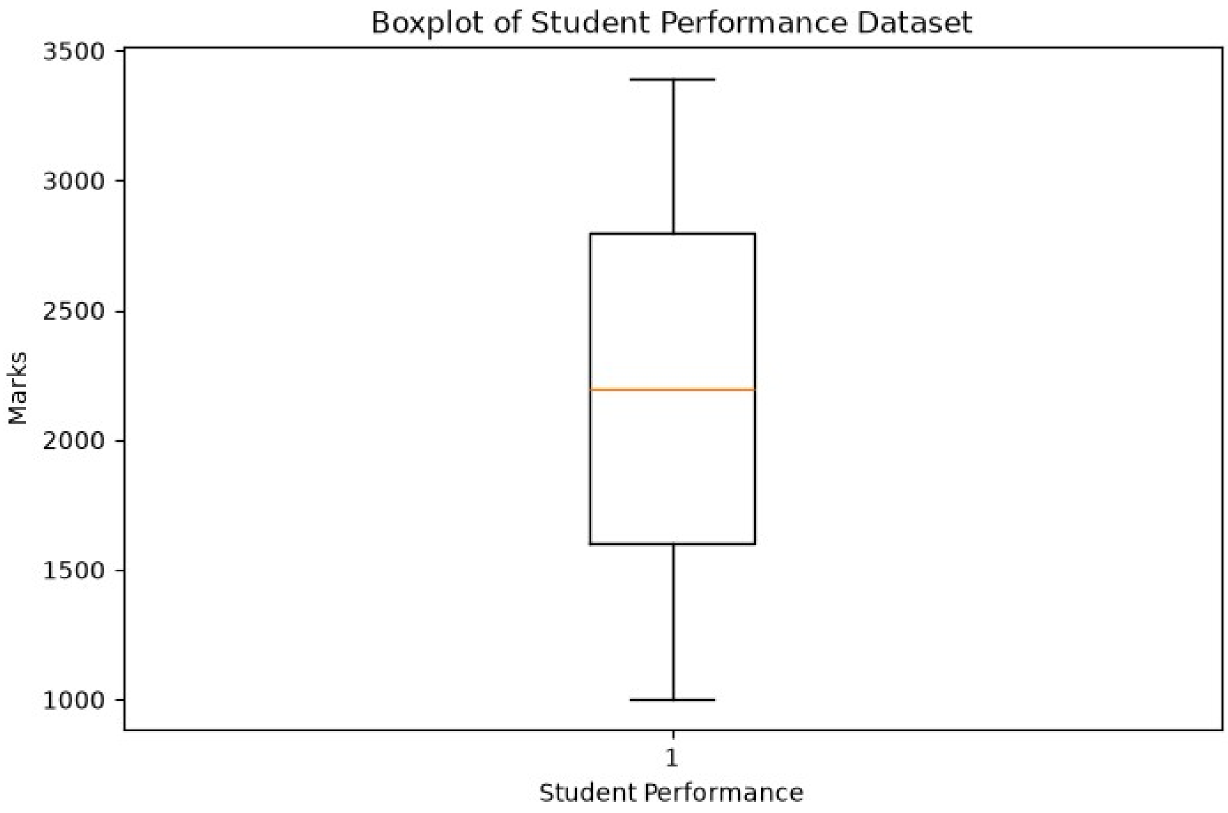
```
1         0         0                1                0       0      0

2        26         0                2                0       0      0

3        14         0                3                1       0      0

4        17         1                3                0       0      0
```

```
   Volunteering       GPA  GradeClass
0             0  2.929196         2.0
1             0  3.042915         1.0
2             0  0.112602         4.0
3             0  2.054218         3.0
4             0  1.288061         4.0
```

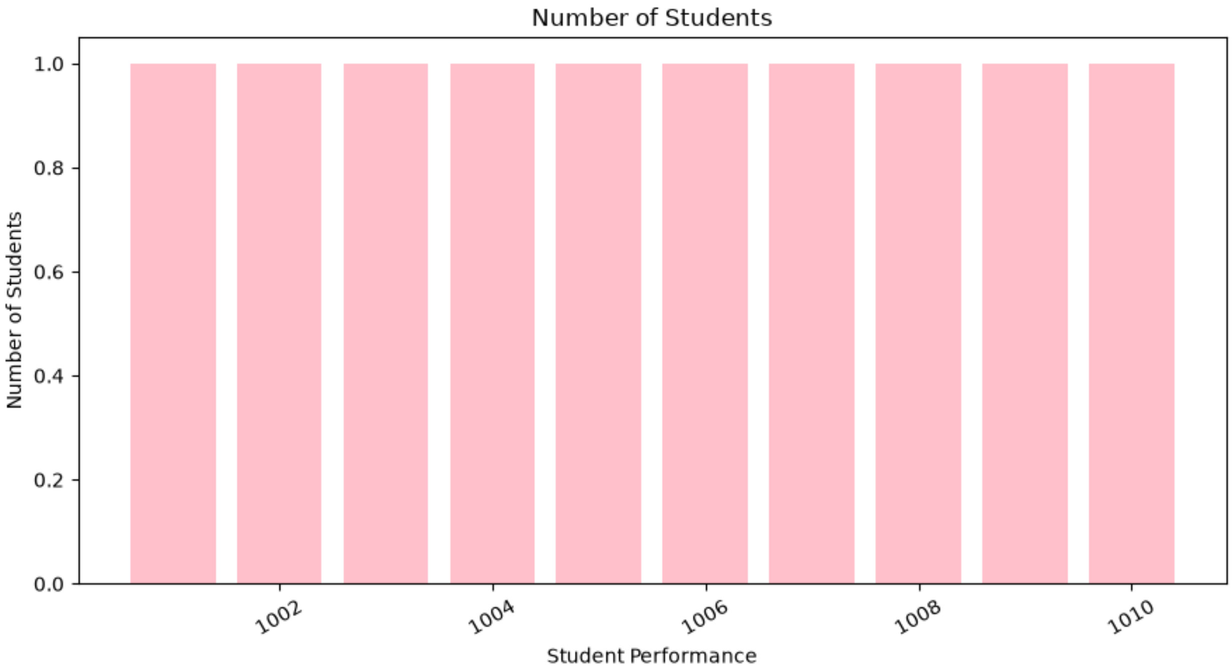
```
import matplotlib.pyplot as plt
import pandas as pd

# Select numerical columns
numeric_cols = student.select_dtypes(include='number')

# Check numerical columns
if numeric_cols.empty:
    print("No numerical columns found!")
else:
    col = numeric_cols.columns[0]

    plt.figure(figsize=(8,5))

    plt.hist(
        student[col].dropna(),
        bins=10,
        color="skyblue",
        edgecolor="black"
    )

    plt.title("Distribution of Student Performance")
    plt.xlabel(col)
    plt.ylabel("Number of Students")

    plt.tight_layout()
    plt.show()
```



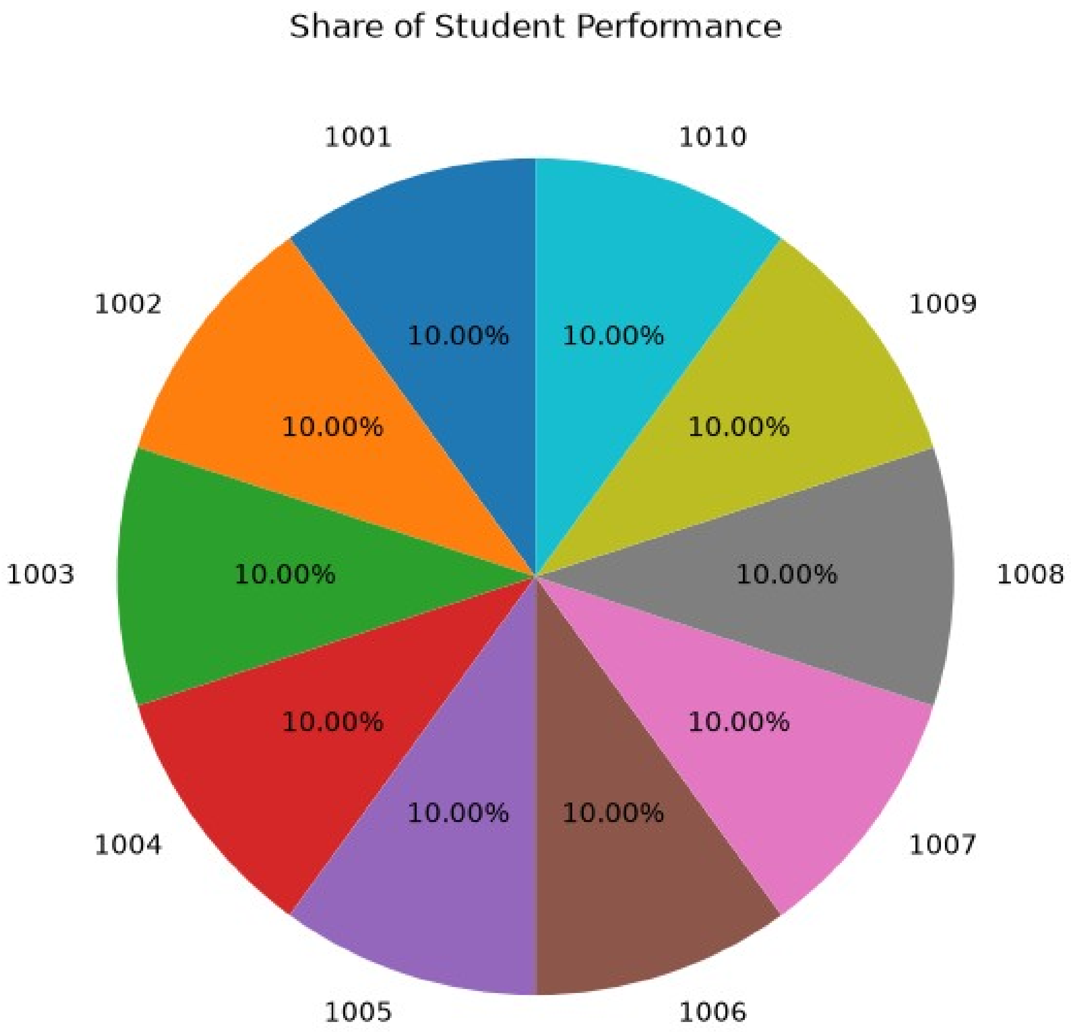
```
import matplotlib.pyplot as plt
import pandas as pd

plt.figure(figsize=(8,5))

plt.boxplot(
    student.select_dtypes(include='number').iloc[:, 0].dropna()
)

plt.title("Boxplot of Student Performance Dataset")
plt.xlabel("Student Performance")
plt.ylabel("Marks")

plt.show()
```



```
import matplotlib.pyplot as plt

# Count the values in the first column
student_counts = student.iloc[:, 0].value_counts().head(10)

# Create Bar Chart
plt.figure(figsize=(9,5))

plt.bar(
    student_counts.index,
    student_counts.values,
    color="pink"
)

plt.title("Number of Students")
plt.xlabel("Student Performance")
plt.ylabel("Number of Students")

plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



```python
import matplotlib.pyplot as plt

# Count the values in the first column
student_counts = student.iloc[:, 0].value_counts().head(10)

# Create Pie Chart
plt.figure(figsize=(7,7))

plt.pie(
    student_counts,
    labels=student_counts.index,
    autopct="%1.2f%%",
    startangle=90
)

plt.title("Share of Student Performance")
plt.show()
```



```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,5))

sns.histplot(
    student.select_dtypes(include='number').iloc[:, 0].dropna(),
    bins=10,
    color="red",
    edgecolor="black",
    kde=True
)

plt.title("Distribution of Student Performance")
plt.xlabel("Student Marks")
plt.ylabel("Number of Students")
plt.tight_layout()
plt.show()
```